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INTRODUCTION TO ETHICAL HACKING

MODULE 01

VYANKATESH ANDIL

1: FOUNDATIONS OF ETHICAL HACKING & INFORMATION SECURITY

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1: FOUNDATIONS OF ETHICAL HACKING & INFORMATION SECURITY

1: Foundations of Ethical Hacking & Information Security

1. INTRODUCTION TO ETHICAL HACKING

Ethical hacking is the **authorized practice of identifying security weaknesses** in computer systems, networks, and applications by simulating the actions of malicious attackers. The primary objective is to **strengthen security posture before real attackers exploit vulnerabilities**.

Ethical hackers operate under **legal permission**, defined **scope**, and **rules of engagement**, distinguishing them from malicious hackers.

2. OBJECTIVES OF ETHICAL HACKING

The main objectives include:

- Identifying security vulnerabilities
- Assessing organizational security posture
- Preventing data breaches
- Ensuring regulatory compliance
- Improving incident response readiness

Ethical hacking answers the question:

“What can an attacker do if they target this system?”

3. INFORMATION SECURITY OVERVIEW

3.1 What Is Information Security?

Information security focuses on **protecting information and information systems** from unauthorized access, disclosure, alteration, and destruction.

3.2 CIA Triad

The foundation of information security is the **CIA Triad**:

Confidentiality

1: FOUNDATIONS OF ETHICAL HACKING & INFORMATION SECURITY

- Prevents unauthorized disclosure of data
- Example: Encryption, access control

Integrity

- Ensures data accuracy and trustworthiness
- Example: Hashing, digital signatures

Availability

- Ensures systems and data are accessible when needed
- Example: Redundancy, backups

4. ETHICAL HACKING VS INFORMATION SECURITY

Ethical Hacking	Information Security
Offensive approach	Defensive approach
Finds vulnerabilities	Prevents attacks
Simulates attackers	Builds protections
Short-term testing	Continuous protection

Ethical hacking is a **subset** of information security.

5. TYPES OF HACKERS

5.1 White Hat Hackers

- Authorized professionals
 - Follow legal contracts
 - Improve security
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5.2 Black Hat Hackers

- Malicious intent

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- Illegal activities
 - Financial or political motivation
-

5.3 Gray Hat Hackers

- No malicious intent
 - May violate laws unintentionally
 - Mixed ethical boundaries
-

5.4 Script Kiddies

- Limited knowledge
 - Use ready-made tools
 - Opportunistic attackers
-

5.5 Hacktivists

- Ideological motivation
 - Political or social agendas
-

5.6 State-Sponsored Hackers

- Government-backed
 - Highly skilled
 - Target national infrastructure
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6. SKILLS REQUIRED FOR AN ETHICAL HACKER

An ethical hacker must understand:

- Networking fundamentals
- Operating systems (Linux, Windows)

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- Programming & scripting
 - Web technologies
 - Cryptography basics
 - Security tools & methodologies
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7. ETHICAL HACKING ENGAGEMENT LIFECYCLE

1. Authorization & scope definition
2. Information gathering
3. Vulnerability analysis
4. Exploitation
5. Post-exploitation
6. Reporting

This lifecycle is explored deeply in later CEH modules.

8. RISKS OF ETHICAL HACKING

If performed incorrectly, ethical hacking may:

- Disrupt services
 - Cause data loss
 - Violate laws
 - Damage reputation
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9. RISK MITIGATION MEASURES

- Written authorization
 - Defined scope
 - Non-destructive testing
-

1: FOUNDATIONS OF ETHICAL HACKING & INFORMATION SECURITY

- Logging & documentation
 - Legal compliance
-

10. CEH Exam-Oriented Notes

- Ethical hacking = **authorized hacking**
- CIA Triad is fundamental
- White hat ≠ Black hat
- Ethical hacking is offensive security
- Permission is mandatory

2: HACKING CONCEPTS, ATTACK SURFACE & THREAT LANDSCAPE

2: Hacking Concepts, Attack Surface & Threat Landscape

1. WHAT IS HACKING?

Hacking refers to the process of **identifying, exploiting, and manipulating vulnerabilities** in systems, networks, or applications to gain **unauthorized or authorized access**, depending on intent and permission.

In ethical hacking, hacking is used as a **controlled security assessment technique** to discover weaknesses before attackers do.

2. WHY SYSTEMS ARE VULNERABLE

No system is 100% secure. Vulnerabilities arise due to:

- Software bugs
- Misconfigurations
- Weak passwords
- Outdated systems
- Poor security policies
- Human errors

Attackers exploit these weaknesses to compromise systems.

3. ATTACK SURFACE

3.1 Definition

The **attack surface** is the total number of points where an attacker can attempt to enter or extract data from a system.

3.2 Types of Attack Surface

Network Attack Surface

- Open ports

2: HACKING CONCEPTS, ATTACK SURFACE & THREAT LANDSCAPE

- Services
- Protocols

Application Attack Surface

- Web forms
- APIs
- Authentication mechanisms

Human Attack Surface

- Social engineering
 - Phishing
 - Insider threats
-

4. THREAT LANDSCAPE

The threat landscape refers to the **constantly evolving ecosystem of cyber threats**, attackers, tools, and techniques.

Modern threats include:

- Advanced Persistent Threats (APTs)
 - Ransomware
 - Supply chain attacks
 - Zero-day exploits
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5. THREAT ACTORS

5.1 Cybercriminals

- Financial motivation
- Data theft, ransomware

2: HACKING CONCEPTS, ATTACK SURFACE & THREAT LANDSCAPE

5.2 Hacktivists

- Political or social causes

5.3 Insider Threats

- Employees or contractors
- Intentional or accidental

5.4 Nation-State Actors

- Cyber warfare
 - Espionage
-

6. COMMON ATTACK VECTORS

- Phishing
- Malware
- Credential theft
- Exploiting vulnerabilities
- Misconfigurations

Attack vectors represent **how attacks are executed**, not the vulnerabilities themselves.

7. VULNERABILITY VS THREAT VS RISK

Term	Definition
Vulnerability	Weakness in system
Threat	Potential attacker
Risk	Probability × impact

Understanding this distinction is **critical for CEH exams**.

2: HACKING CONCEPTS, ATTACK SURFACE & THREAT LANDSCAPE

8. REAL-WORLD ATTACK EXAMPLE

Case: Data Breach via Misconfiguration

- Cloud storage left public
- No authentication
- Result: Data leakage

Lesson: Configuration errors are major attack vectors.

9. ROLE OF ETHICAL HACKER IN THREAT MANAGEMENT

Ethical hackers:

- Identify attack surfaces
- Simulate threat actors
- Reduce organizational risk
- Improve defenses

10. MITRE ATT&CK

Phase	MITRE Tactic
Reconnaissance	TA0043
Initial Access	TA0001
Execution	TA0002

11. CEH Exam-Oriented Notes

- Attack surface = entry points
- Threat ≠ vulnerability
- Risk = likelihood × impact
- Phishing is an attack vector
- Misconfiguration is a major risk

3: ETHICAL HACKING LAWS, LEGAL FRAMEWORK & COMPLIANCE

3: Ethical Hacking Laws, Legal Framework & Compliance

1. IMPORTANCE OF LEGAL KNOWLEDGE IN ETHICAL HACKING

Ethical hacking is **only ethical and legal when performed with proper authorization**. Without legal permission, the same technical actions used in ethical hacking are considered **criminal offenses**.

For CEH professionals, understanding legal boundaries is as important as technical skills.

2. AUTHORIZATION IN ETHICAL HACKING

2.1 Written Permission

Before performing any security assessment, an ethical hacker must obtain:

- Written authorization
- Signed contract or agreement
- Clearly defined scope

This authorization protects both:

- The organization
 - The ethical hacker
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2.2 Scope Definition

Scope defines:

- Systems to be tested
- Testing methods allowed
- Time window
- Data handling rules

Testing outside scope = **illegal activity**.

3: ETHICAL HACKING LAWS, LEGAL FRAMEWORK & COMPLIANCE

3. CYBER LAWS AND REGULATIONS

3.1 Computer Fraud and Abuse Act (CFAA – USA)

- Prohibits unauthorized access to computer systems
- Severe penalties for violations

3.2 Information Technology Act (IT Act – India)

- Governs cybercrime and electronic data
- Section 43 & 66: penalties for unauthorized access

3.3 General Data Protection Regulation (GDPR – EU)

- Protects personal data
- Requires data security and breach reporting

3.4 Other Global Cyber Laws

- UK Computer Misuse Act
 - Australian Cybercrime Act
-

4. ETHICAL BOUNDARIES IN HACKING

Ethical hackers must:

- Respect privacy
- Avoid unnecessary data access
- Report findings responsibly
- Avoid data modification unless permitted

5. LEGAL CONSEQUENCES OF UNAUTHORIZED HACKING

Unauthorized hacking may lead to:

- Criminal charges
- Heavy fines
- Imprisonment
- Career damage

3: ETHICAL HACKING LAWS, LEGAL FRAMEWORK & COMPLIANCE

6. COMPLIANCE & SECURITY STANDARDS

6.1 ISO/IEC 27001

- Information security management system

6.2 PCI-DSS

- Payment card data protection

6.3 HIPAA

- Healthcare data security

Ethical hacking helps organizations **meet compliance requirements**.

7. RULES OF ENGAGEMENT (ROE)

RoE defines:

- Testing approach
 - Escalation procedures
 - Incident handling
 - Reporting format
-

8. ETHICAL HACKING CODE OF CONDUCT

CEH professionals must:

- Maintain confidentiality
- Avoid conflict of interest
- Follow professional ethics

9. RISK MITIGATION DURING LEGAL ENGAGEMENTS

- Use non-destructive tests
- Maintain logs
- Immediate reporting of critical issues

3: ETHICAL HACKING LAWS, LEGAL FRAMEWORK & COMPLIANCE

10. CEH Exam-Oriented Notes (Part 3)

- Authorization is mandatory
- Scope defines legality
- IT Act Section 66 (India) is important
- GDPR protects personal data
- Ethical ≠ illegal

4: ETHICAL HACKING METHODOLOGIES & FRAMEWORKS

4: Ethical Hacking Methodologies & Frameworks

1. IMPORTANCE OF METHODOLOGIES IN ETHICAL HACKING

A methodology provides a **systematic, repeatable, and controlled approach** to performing ethical hacking. Without a structured methodology, security testing can be incomplete, risky, or legally unsafe.

CEH strongly emphasizes **methodology-driven hacking** rather than random tool usage.

2. ETHICAL HACKING METHODOLOGY

Ethical hacking generally follows these phases:

1. Reconnaissance
2. Scanning & Enumeration
3. Gaining Access
4. Maintaining Access
5. Covering Tracks
6. Reporting

These phases simulate **real-world attacker behavior**.

3. RECONNAISSANCE (FOOTPRINTING)

3.1 Definition

Reconnaissance involves **gathering information about the target** without direct interaction (passive) or with limited interaction (active).

3.2 Objectives

- Identify IP ranges
- Domain details
- Technology stack
- Employees and emails

4: ETHICAL HACKING METHODOLOGIES & FRAMEWORKS

3.3 Importance

Better reconnaissance = easier exploitation.

4. SCANNING & ENUMERATION

4.1 Scanning

- Identifies live systems
 - Open ports
 - Services
-

4.2 Enumeration

- Extracts deeper information
 - Usernames
 - Network shares
-

5. GAINING ACCESS (EXPLOITATION)

This phase involves:

- Exploiting vulnerabilities
- Password attacks
- Malware injection

Goal: **Obtain unauthorized access within scope.**

6. MAINTAINING ACCESS (PERSISTENCE)

Attackers attempt to:

- Install backdoors
- Maintain long-term access

Ethical hackers **simulate** this phase without causing harm.

4: ETHICAL HACKING METHODOLOGIES & FRAMEWORKS

7. COVERING TRACKS

Attackers erase logs and traces to avoid detection.

Ethical hackers:

- Demonstrate the risk
 - Do not actually destroy logs
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8. REPORTING (MOST CRITICAL PHASE)

Includes:

- Vulnerability details
 - Impact analysis
 - Proof of concept
 - Remediation steps
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9. CYBER KILL CHAIN FRAMEWORK

Developed by Lockheed Martin:

1. Reconnaissance
2. Weaponization
3. Delivery
4. Exploitation
5. Installation
6. Command & Control
7. Actions on Objectives

10. MITRE ATT&CK FRAMEWORK

MITRE ATT&CK provides:

4: ETHICAL HACKING METHODOLOGIES & FRAMEWORKS

- Tactics
- Techniques
- Real-world adversary behavior

Used for:

- Threat modeling
- Detection mapping

11. COMPARISON: CEH METHODOLOGY VS MITRE

CEH Phase	MITRE Tactic
Reconnaissance	Reconnaissance
Gaining Access	Initial Access
Persistence	Persistence
Covering Tracks	Defense Evasion

12. CEH Exam-Oriented Notes (Part 4)

- Methodology ensures completeness
- Reporting is mandatory
- Kill chain explains attack flow
- MITRE focuses on adversary behavior
- Ethical hackers simulate attacks

5: PENETRATION TESTING, VULNERABILITY ASSESSMENT & SECURITY TESTING TYPES

5: Penetration Testing, Vulnerability Assessment & Security Testing Types

1. UNDERSTANDING SECURITY ASSESSMENTS

Organizations use multiple security testing approaches to identify weaknesses. Ethical hacking is often confused with penetration testing and vulnerability assessment, but each serves a **different purpose**.

2. VULNERABILITY ASSESSMENT (VA)

2.1 Definition

Vulnerability Assessment is the process of:

- Identifying
- Classifying
- Prioritizing vulnerabilities

It focuses on **discovery**, not exploitation.

2.2 Objectives

- Identify known vulnerabilities
- Assess exposure level
- Provide risk ranking

2.3 Characteristics

- Mostly automated
- Uses scanners
- No exploitation

2.4 Limitations

- Cannot confirm exploitability
- False positives possible

5: PENETRATION TESTING, VULNERABILITY ASSESSMENT & SECURITY TESTING TYPES

3. PENETRATION TESTING (PT)

3.1 Definition

Penetration testing involves **actively exploiting vulnerabilities** to determine the real-world impact of a security weakness.

3.2 Objectives

- Validate vulnerabilities
 - Demonstrate business impact
 - Test detection & response
-

3.3 Types of Penetration Testing

Black Box Testing

- No prior knowledge
- Simulates external attacker

White Box Testing

- Full knowledge
- Source code, architecture provided

Gray Box Testing

- Partial knowledge
- Most realistic approach

5: PENETRATION TESTING, VULNERABILITY ASSESSMENT & SECURITY TESTING TYPES

4. ETHICAL HACKING VS PENETRATION TESTING VS VA

Aspect	VA	PT	Ethical Hacking
Exploitation	✗	✓	✓
Authorization	Limited	Required	Required
Scope	Broad	Targeted	Broad
Goal	Find flaws	Prove impact	Improve security

5. RED TEAM VS BLUE TEAM

5.1 Red Team

- Offensive security
- Simulates real attackers

5.2 Blue Team

- Defensive security
- Detection and response

5.3 Purple Team

- Collaboration between Red & Blue

6. AUTOMATED VS MANUAL TESTING

- Automated tools = speed
- Manual testing = accuracy

Best practice = **hybrid approach**.

5: PENETRATION TESTING, VULNERABILITY ASSESSMENT & SECURITY TESTING TYPES

7. SECURITY TESTING TOOLS (HIGH-LEVEL)

- Vulnerability scanners
- Exploitation frameworks
- Network scanners
- Web testing tools

(Detailed tools are covered in later modules.)

8. RISK & BUSINESS IMPACT

Security findings should be reported in terms of:

- Confidentiality impact
 - Integrity impact
 - Availability impact
-

9. REPORTING & REMEDIATION

Reports must include:

- Severity rating
 - Proof of concept
 - Remediation steps
-

10. CEH Exam-Oriented Notes (Part 5)

- VA ≠ PT
- VA finds, PT proves
- Black box = no knowledge
- Ethical hacking is broader
- Purple team = collaboration

CEH CERTIFICATION, CAREER PATHS, ETHICS, SUMMARY & EXAM STRATEGY

CEH Certification, Career Paths, Ethics, Summary & Exam Strategy

1. CEH CERTIFICATION OVERVIEW

Certified Ethical Hacker (CEH) is a globally recognized certification by **EC-Council** that validates an individual's ability to:

- Identify vulnerabilities
- Use ethical hacking tools
- Follow legal and ethical standards

CEH focuses on **attacker mindset with defensive intent**.

2. IMPORTANCE OF CEH IN CYBERSECURITY

CEH bridges the gap between:

- Theory and practice
- Defensive security and offensive testing

It prepares professionals to think like attackers while acting ethically.

3. ETHICAL HACKING CAREER PATHS

3.1 Ethical Hacker / Penetration Tester

- Performs security assessments
- Identifies and exploits vulnerabilities

3.2 Security Analyst

- Monitors threats
- Responds to incidents

CEH CERTIFICATION, CAREER PATHS, ETHICS, SUMMARY & EXAM STRATEGY

3.3 SOC Analyst

- Handles alerts
- Performs threat analysis

3.4 Red Team Member

- Simulates real-world attacks

3.5 Cybersecurity Consultant

- Advises organizations on security posture
-

4. SKILLS DEVELOPED THROUGH CEH

- Networking & system knowledge
 - Web application security
 - Cryptography basics
 - Security tools proficiency
 - Reporting and communication
-

5. PROFESSIONAL ETHICS IN ETHICAL HACKING

Ethical hackers must:

- Maintain confidentiality
- Avoid misuse of access
- Follow scope and authorization
- Disclose vulnerabilities responsibly

Ethics defines the **difference between hacker and criminal**.

6. ETHICAL HACKING IN REAL ORGANIZATIONS

Organizations use ethical hacking to:

- Reduce breach risk
-

CEH CERTIFICATION, CAREER PATHS, ETHICS, SUMMARY & EXAM STRATEGY

- Comply with regulations
- Improve security maturity

Ethical hacking is now a **core cybersecurity function**.

7. SUMMARY OF MODULE 01

This module introduced:

- Ethical hacking concepts
- Types of hackers
- Legal frameworks
- Methodologies
- Security assessments
- Career opportunities

It forms the **foundation of all CEH modules**.

8. CEH EXAM STRATEGY

8.1 High-Weight Topics

- CIA Triad
 - Types of hackers
 - Legal authorization
 - VA vs PT vs Ethical Hacking
 - Hacking phases
-

8.2 Common Exam Mistakes

- Ignoring legality
- Confusing VA and PT

CEH CERTIFICATION, CAREER PATHS, ETHICS, SUMMARY & EXAM STRATEGY

- Forgetting scope importance

9. QUICK REVISION TABLE

Concept	Key Point
Ethical Hacking	Authorized hacking
CIA Triad	Confidentiality, Integrity, Availability
White Hat	Legal hacker
VA	Identify vulnerabilities
PT	Exploit vulnerabilities

10. CONCLUSION

Ethical hacking is a **disciplined, legal, and structured approach** to cybersecurity testing. CEHv13 Module 01 establishes the ethical, legal, and methodological foundation required to responsibly perform advanced hacking techniques covered in later modules.

Mastering this module ensures:

- Legal safety
- Ethical professionalism
- Strong conceptual clarity